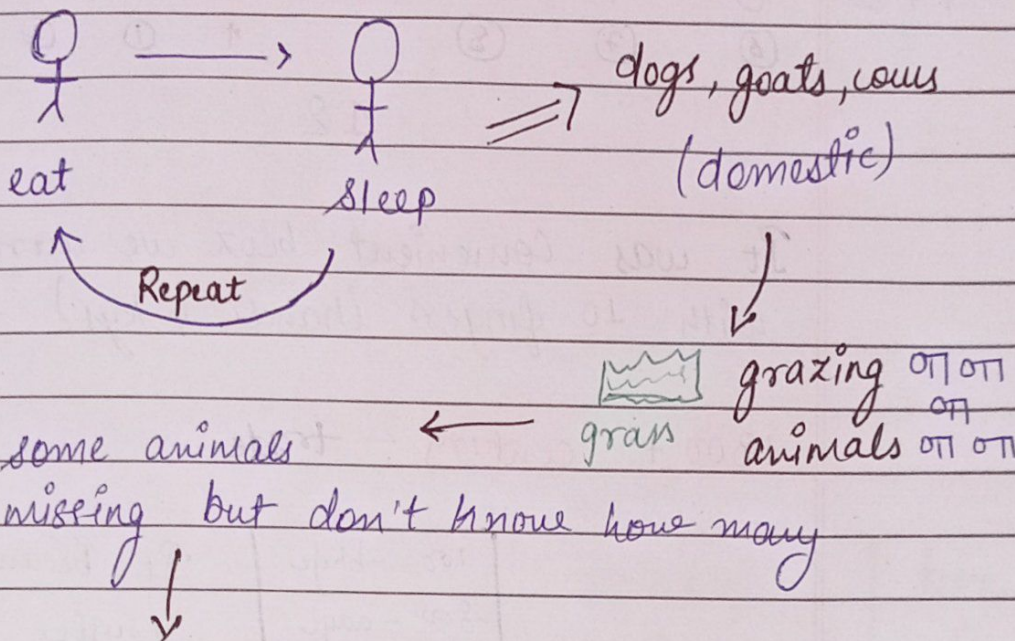


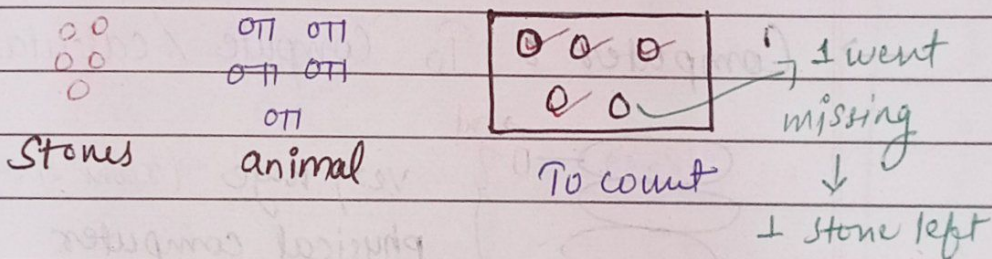
Lec 1 Basic of Internet

Evolution of Computer

Date 16/09/24
Page _____



Tally mark (first system to count)



but what for 1000?

so we needed number system to express in language

Egypt (base 60) — 60 units } not efficient

Then base 10 (india)

[0 1 2 3 4 5 6 7 8 9]

7+4 0 1 2 3 4 5 6 7 8 9
 ③ ④ ① ②

beoz we took 1 round
11

4 + 8

0 1 2 3 4 5 6 7 8 9
(6) (7) (8) ↑ (1) (2) (3) (4) (5)

12

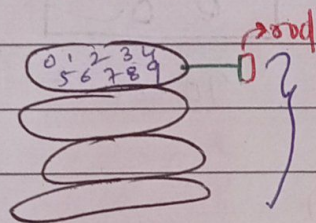
It was convenient bcoz we were familiar with 10 fingers (hands & legs)

1800 + century - trade

100	- bhiye
800	- aaye
190	- bheje

It became tough to calculate

→ Computer : To Compute / calculate

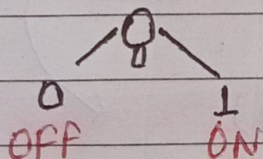


very huge (room size)
physical computer
by rotating rod

Accurate but slow

Then we got transistors

It is like bulb.



Binary number system
(base 2)

decimal

0	0 + 1 = 1
1	0 + 0 = 0
1	1 + 0 = 1
2	1 + 1 = 10

0 1
binary

1+1

0 1
1 ↑

we make a round

1 0

10

1 1

+ 1

+ 1

11 (3)

100 (4)

Decimal

Binary

hex

0

0

0

1

1

1

2

10

2

3

11

3

4

100

4

5

101

5

6

110

6

7

111

7

8

1000

8

9

1001

9

10

1010

A

11

1011

B

12

1100

C

13

1101

D

14

1110

E

15

1111

F

Further

decimal $\rightarrow$ binary

★

27 $\rightarrow$ binary

base	2	27		$\rightarrow$ rem
of	2	13	1	
binary	2	6	1	
	2	3	0	
	2	1	1	
		0	1	

$\Rightarrow 11011$

★

binary $\rightarrow$ decimal

decimal $\rightarrow$
decimal

$\rightarrow 386 = 3 \times 10^2 + 8 \times 10^1 + 6 \times 10^0$

$\Downarrow$

386

$11011 = 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$

$8 + 4 + 0 + 1$

13

$\rightarrow$ Octadecimal { 0, 1, 2, 3, 4, 5, 6, 7 }

8	37	
8	4	5
	0	4

45 - octadecimal of 37

$45 = 4 \times 8^1 + 5 \times 8^0$

$= 32 + 5$

$=$ 37 decimal of 45 - octadecimal

16

→ Hexadecimal { 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F }
(10 is not unique)

16	38	
16	2	6
	0	2

$$\begin{aligned} 26 &= 2 \times 16^1 + 6 \times 16^0 \\ &= 32 + 6 \\ &= 38 \end{aligned}$$

Confusion

hexadecimal

10 → decimal

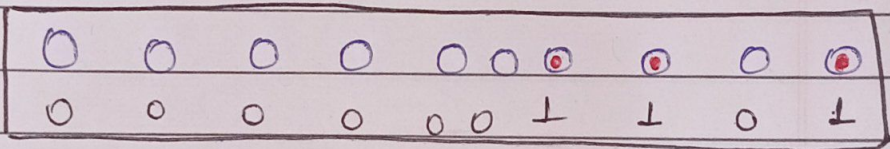
$$\begin{aligned} 10 &= 10 \times 16^0 \\ &= 10 \end{aligned}$$

or

$$\begin{aligned} 10 &= 1 \times 16^1 + 0 \times 16^0 \\ &= 16 \end{aligned}$$

→ Transistor

To store 13 → 1101

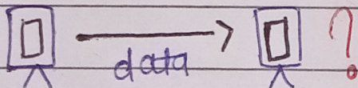


13

That's why binary is used to store numbers by computer.

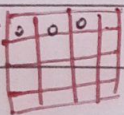
computer system came in 1950's

1950



cd, pen drive, HDD

punch sheet by



1110 sheet

transfer by

very complex when transfer b/w diff. cities.